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h
6.62607015×
10-34 J.
λ

```

$$E_{\text{photon}} = \frac{h c}{\lambda},$$

(1)

$$R_{\text{dark}}(T) \propto \exp\left(-\frac{E_g}{2 k_B T}\right),$$

(2)

$$H(X) = -\sum_x p_x \log_2 p_x$$

$$p_x \rightarrow$$

$$\log(0)$$

$$\text{ENTROPY_PROB_CLAMP}$$

$$10 \varepsilon_{\text{mach}}$$

$$\text{numpy.finfo(float).eps}$$

$$2.22 \times 10^{-16}$$

$$\text{ENTROPY_PROB_CLAMP} = 10 \cdot \varepsilon_{\text{mach}} \approx 2.22 \times 10^{-15}.$$

(3)

```

clamp_probability
clamp_probabilities
[δ, 1-
δ]
δ =
ENTROPY_PROB_CLAMP
Y1_SAFE_THRESHOLD
max(10-12, 100 εmach)
Y1
R ≥
q{ -Qμf(Eμ)H2(Eμ)+
Q1[1-
H2(e1)]}
Q1
e1
Y1
Y1
H2
Y1_SAFE_THRESHOLD
Y1
NUMERIC_ABS_TOL
=
10-12
NUMERIC_REL_TOL
=
10-9
PROB_SUM_TOL

```